

EDUCATION

- **Radboud University** Nijmegen, The Netherlands
PhD position in: Calculus of Variations; *Supervisor:* Riccardo Cristoferi
September 2023 - August 2027
Area of interest: Calculus of variations, geometric measure theory, Gamma-convergence, material science and related fields.
- **Polytechnic University of Turin** Turin, Italy
Master degree in: Engineering Mathematics
September 2021 - July 2023
Thesis: Anti-plane analysis of Francfort-Marigo model for quasi-static brittle fracture; *Supervisor:* Marco Morandotti.
- **KTH Royal Institute of Technology** Stockholm, Sweden
Exchange semester
August 2022 - January 2023
- **Polytechnic University of Turin** Turin, Italy
Bachelor degree in: Mathematics for Engineering
September 2018 - July 2021
Thesis: Fractal dimension and Hausdorff measure; *Supervisor:* Luca Lussardi
- **L.S. Battaglini** Taranto, Italy
Scientific Diploma
September 2013 - July 2018

PUBLICATIONS

- **Low regularity potentials in heterogeneous Cahn–Hilliard functionals**
R. Cristoferi, J. Deutsch, L. Pignatelli
Arxiv: 2603.25369
- **Phase separation in multiply periodic materials with fine microstructures**
R. Cristoferi, L. Pignatelli
Arxiv: 2506.14462
- **Phase separation in multiply periodic materials with fine microstructures: a second regime**
L. Pignatelli
ongoing
- **Sharp interface dynamics in heterogeneous domains**
R. Cristoferi, L. Gancedi, L. Pignatelli
ongoing
- **Convergence of critical regime to sub- and super-critical regimes**
R. Cristoferi, L. Pignatelli
ongoing

TALKS

- **A Taxonomy of Separation: from Flory-Huggins to non-local gradients**
06-07-2026, Workshop Calculus of Variations NL
Roermond
- **Sharp interface limit of the Allen-Cahn equation with small periodic heterogeneities**
08-05-2026, Seminar on Calculus of Variations
TU Wien, Vienna
- **Why Soap Films are smarter than Calculus**
16-12-2025, PhD Colloquium
Radboud University, Nijmegen
- **Phase separation in multiply periodic materials with fine microstructures**
10-07-2025, Workshop Calculus of Variations NL
Roermond
- **Phase separation in heterogenous media**
28-05-2025, Seminar on Calculus of Variations
TU Wien, Vienna
- **Homogenization and phase separation in heterogeneous media**
20-03-2025, Seminar talk
RWTH Aachen University, Aachen
- **Gamma Mia - an introduction to Gamma convergence**
14-10-2024, PhD Colloquium
Radboud University, Nijmegen

- **Quasi-static brittle fracture: analysis of Francfort-Marigo model**

22-11-2023, Applied Analysis Seminar

Radboud University, Nijmegen

POSTERS

- **Sharp interface limit of the Allen–Cahn equation with small periodic heterogeneities**

11-06-2026, Calculus of Variations and Applications to the Real World

INdAM Meeting, Cortona

- **Phase separation in heterogeneous media**

12-06-2025, Mathematical Analysis and Applications

Summer School, Como

- **Phase separation in multi-scale media**

11-04-2024, Mathematical Advances for Complex Materials with Microstructures

University of Würzburg, Würzburg

RESEARCH VISITS

- **TU Wien, Vienna**

Collaboration with Jakob Deutsch

04-05-2026 - 08-05-2026

- **TU Wien, Vienna**

Collaboration with Jakob Deutsch

26-05-2025 - 30-05-2025

- **RWTH Aachen University, Aachen**

Collaboration with Likhit Ganedi

17-03-2025 - 20-03-2025

PROJECTS

- **Reading Group organizer**

Series of meetings for the CalcVarNL community - Main topics: Gradient Flows, Optimal Transport, Stochastic Analysis.

The Netherlands, *September 2023 - ongoing*

- **Modeling Camp organizer**

Collaboration with dutch companies, where problems from the industry are modeled mathematically and numerically.

Radboud University, *February 2024, February 2025*

- **Head of YSC committee**

Young Scientist Council of the Institute for Mathematics, Astrophysics and Particle Physics.

Radboud University, *March 2024 - ongoing*

EVENTS ATTENDED

- **Workshop Calculus of Variations NL**

06-07-2026 - 10-07-2026

Roermond, The Netherlands

- **Calculus of Variations and Applications to the Real World**

08-06-2026 - 12-06-2026

INdAM Meeting

Cortona, Italy

- **Modern Methods in Calculus of Variations**

23-02-2026 - 27-02-2026

Hausdorff Center for Mathematics

Bonn, Germany

- **Workshop Calculus of Variations NL**

07-07-2025 - 11-07-2025

Roermond, The Netherlands

- **Mathematical Analysis and Applications**

09-06-2025 - 13-06-2025

Summer School

Como, Italy

- **Analysis of Partial Differential Equations**

16-09-2024 - 19-09-2024

Czech Academy of Sciences

Prague, Czech Republic

- **Workshop Calculus of Variations NL**

08-07-2024 - 12-07-2024

Schiermonnikoog, The Netherlands

- **Vito Volterra Meeting in Calculus of Variations 2024**

24-06-2024 - 28-06-2024

Sapienza University of Rome

Rome, Italy

- **Mathematical Advances for Complex Materials with Microstructures**

08-04-2024 - 12-04-2024

University of Würzburg

Würzburg, Germany

- **Variational models for material failure**

20-09-2023 - 22-09-2023

FAU Erlangen-Nürnberg

Erlangen, Germany

- **Analysis of Partial Differential Equations**

18-09-2023 - 20-09-2023

KU Eichstätt-Ingolstadt

Eichstätt, Germany

- **Beyond Elasticity: Advances and Research Challenges**

16-05-2022 - 20-05-2022

CIRM

Luminy, France

TEACHING

- **Numerical Methods for PDEs**
TA for the bachelor course Numerical Methods for PDEs
February 2026 - June 2026
prof. V. Nikolić
- **Numerical Methods**
TA for the bachelor course Numerical Methods
September 2025 - January 2026
prof. L. Scarabosio
- **Calculus of Variations**
TA for the MasterMath course Calculus of Variations
February 2025 - June 2025
prof. A. Bach, prof. R. Cristoferi
- **Multivariable Analysis**
TA for the bachelor course Multivariable Analysis
September 2024 - January 2025
prof. R. Cristoferi
- **Numerical Methods for PDEs**
TA for the bachelor course Numerical Methods for PDEs
February 2024 - June 2024
prof. V. Nikolić
- **Analysis 2**
TA for the bachelor course Analysis 2
September 2023 - January 2024
prof. R. Cristoferi

SKILLS

- **Languages:** Italian (native), English (fluent), Dutch (basic), Spanish (intermediate)
- **Coding:** C++, Python, Matlab, $\text{\LaTeX}$ – all advanced

ACADEMIC REFEREES

- **Riccardo Cristoferi:** Radboud University, cristoferi@science.ru.nl
- **Marco Morandotti:** Polytechnic University of Turin, marco.morandotti@polito.it

Luca Pignatelli
Radboud University, Nijmegen
29-06-2026